

Generated Question Paper

Q1. State two advantages of using a pyramid of energy rather than a pyramid of biomass to represent a food chain. [2 marks]

Q2. (a) (i) It is estimated that microplastics make up 5 Calculate the mass of microplastics used on one day in 2013 in these countries. Show working and units. [3 marks]

Q3. (b) Lugworms live in sand on beaches and feed on diatoms, which require ammonium ions. Researchers tested effects of microplastics on lugworms; see Table 3.1. Explain and explain why researchers concluded that non-biodegradable microplastics are most harmful to lugworms using Table 3.1. [4 marks]

Q4. (ii) Fishing can be managed to promote the conservation of fish stocks. Explain one way that fishing can be managed to conserve fish stocks. [2 marks]

Q5. (b) Fig. 6.2 shows a reproduction method used by some potato farmers. begincenter endcenter Describe the advantages of this type of reproduction in crop production. [3 marks]

Q6. (ii) State the name of the process occurring at step 3 that causes gas bubbles to form in the dough. [1 marks]

Q7. (d) State the names of two products of biotechnology, other than bread, that make use of microorganisms. [2 marks]

Q8. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]

Q9. (a) Describe how the structure of a neurone relates to its function. [3 marks]

Q10. (b) Complete Fig. 5.1 by stating two other organs that belong to two organ systems. begincenter endcenter noindent (c) Fig. 5.2 shows photomicrograph of part of mammalian testis. begincenter endcenter Explain why meiosis is necessary in the testes. [3 marks]

Q11. (d) shows a photomicrograph of a sperm cell. begincenter endcenter noindent (e) Draw and label one human egg cell, including at least one feature not found in a sperm cell. [3 marks]

Q12. (b) Soybean plants, *Glycine max*, were grown in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Fig. 2.1 shows average photosynthesis rates. begincenter endcenter noindent Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

Q13. (e) Explain the effects of nitrate ion deficiency on plant growth. [3 marks]

Q14. (iii) State the name of the gas that forms to create the gas bubbles in the dough. [1 marks]

Q15. (a) On Fig.2, label with an R the position of the right ventricle. [1 marks]

Q16. (c) Explain the reasons for the different temperatures used in step 3 and step 5. [2 marks]

Q17. (a) State the name of organism A in the figure above. [1 marks]

Q18. (d) Scientists had higher breathing rates in the 550 ppm CO₂ plot. [2 marks]

Q19. (f) Describe what happens to a fertilised egg cell before implantation in the uterus. [3 marks]